

7 Animals and plants can respond to light.

(a) The photograph shows the front of a human eye.



(Source: © SCIENCE PHOTO LIBRARY)

(i) The size of the pupil changes when moving from dim to very bright light.

Explain how this change occurs.

(3)

(ii) Describe the role of rhodopsin in causing changes in the polarisation of rod cells.

(4)

*(b) Exposure to light controls the germination of seeds and the direction of growth of plants.

In an investigation the seeds of plantains were exposed to light.

The seeds were first exposed to red light and then to far-red light.

The percentage of seeds that germinated when exposed to different periods of red and far-red light was recorded.

The results are shown in the table.

Exposure to red light / min	Percentage germination (%)						
	Exposure to far-red light /seconds						
	1	5	10	15	30	60	960
32	100	20	14	6	4	6	10
16	92	30	40	28	30	33	20
8	100	62	50	57	44	47	56
4	100	67	60	57	40	37	92

Comment on the effect of red and far-red light on the germination of these seeds and the processes that occur in their cells.

Use the information in the table to support your answer.

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(Total for Question 7 = 13 marks)